ECON 0100 | HW E1 | Costs

Solutions

Answers and work are shown in the red blocks.

Q1 | Skyline Custom Brooms Production

The table shows how many workers it takes Skyline to make each number of brooms by hand. Workers are paid 240 coins a week. There are no fixed costs.

Workers	**0**	**1**	**2**	**3**	**4**
Brooms	0	8	14	18	20

a) What is the marginal product of the second worker?

The second worker adds $14 - 8 = 6$ brooms.

b) What is the marginal product of the fourth worker?

The fourth worker adds $20 - 18 = 2$ brooms.

c) As Skyline adds workers, does the marginal product rise, stay the same, or fall?

It falls: 8, 6, 4, 2. This is why each additional broom takes more time and effort, and why Skyline's marginal cost increases.

d) What is the variable cost of making 14 brooms a week?

14 brooms take 2 workers, so variable cost is $2 \times 240 = 480$ coins.

e) What is the total cost of making 14 brooms a week?

With no fixed cost, total cost is also 480 coins.

Q2 | CloudSweep Brooms Production

The table shows how many workers it takes CloudSweep to make each number of brooms with standardized methods. Workers are paid 60 coins a week. Fixed costs are 1,000 coins a week.

Workers	**0**	**1**	**2**	**3**	**4**
Brooms	0	4	8	12	16

a) What is the marginal product of the second worker?

The second worker adds $8 - 4 = 4$ brooms.

b) What is the marginal product of the fourth worker?

The fourth worker adds $16 - 12 = 4$ brooms.

c) As CloudSweep adds workers, does the marginal product rise, stay the same, or fall?

It stays the same: every worker adds 4 brooms. This is why CloudSweep's marginal cost is constant. Each broom takes a quarter of a worker's week, $60/4 = 15$ coins, which is the flat MC in Graph A1 of Q3.

d) What is the variable cost of making 8 brooms a week?

8 brooms take 2 workers, so variable cost is $2 \times 60 = 120$ coins.

e) What is the total cost of making 8 brooms a week?

Total cost is the fixed cost plus the variable cost: $1,000 + 120 = 1,120$ coins.

Q3 | Cost Structures

The Department of Magical Transportation is studying the broomstick manufacturing industry. Three major manufacturers dominate the market, each with distinct cost structures.

CloudSweep is an established manufacturer known for large facilities and expensive equipment (high fixed costs) and standardized production methods (constant marginal costs).

Skyline Custom manufactures personalized racing brooms from a basic workshop (no fixed costs) with each additional broom taking extra effort to make (increasing marginal costs).

SwiftTech is a new company using automated equipment it rents by the broom (no fixed costs) and standardized production steps (constant marginal costs).

a) Match curve in Row 1 of cost curve matrix to its manufacturer.

Column A: CloudSweep Column B: Skyline Column C: SwiftTech

Column A (Graph A1) is CloudSweep. MC is flat (constant marginal cost) and ATC sits far above it at low output, falling toward it as the high fixed cost is spread over more brooms.

Column B is Skyline. MC slopes up (increasing marginal cost). With no fixed cost, ATC starts beside MC and rises more slowly, because the average includes all the cheaper early brooms.

Column C is SwiftTech. With no fixed cost and a constant marginal cost, every broom costs the same, so $ATC = MC$ and the two curves lie on one flat line.

b) Per Unit Subsidy

The Ministry gives firms 4 coins for each broom they make. Which graph shows SwiftTech's new cost curves?

Graph C1 Graph C2 Graph C3 Graph C4

Graph C4. A per unit subsidy lowers the cost of every broom by 4, so MC and ATC both shift down by 4, from 12 to 8.

c) Per Unit Tax

Which graph shows Skyline's cost curves after the Ministry imposes a 5 coin tax on each broom produced?

Graph B1 Graph B2 Graph B3 Graph B4

Graph B3. A per unit tax raises the cost of every broom by 5, so MC and ATC both shift up by 5 at every quantity.

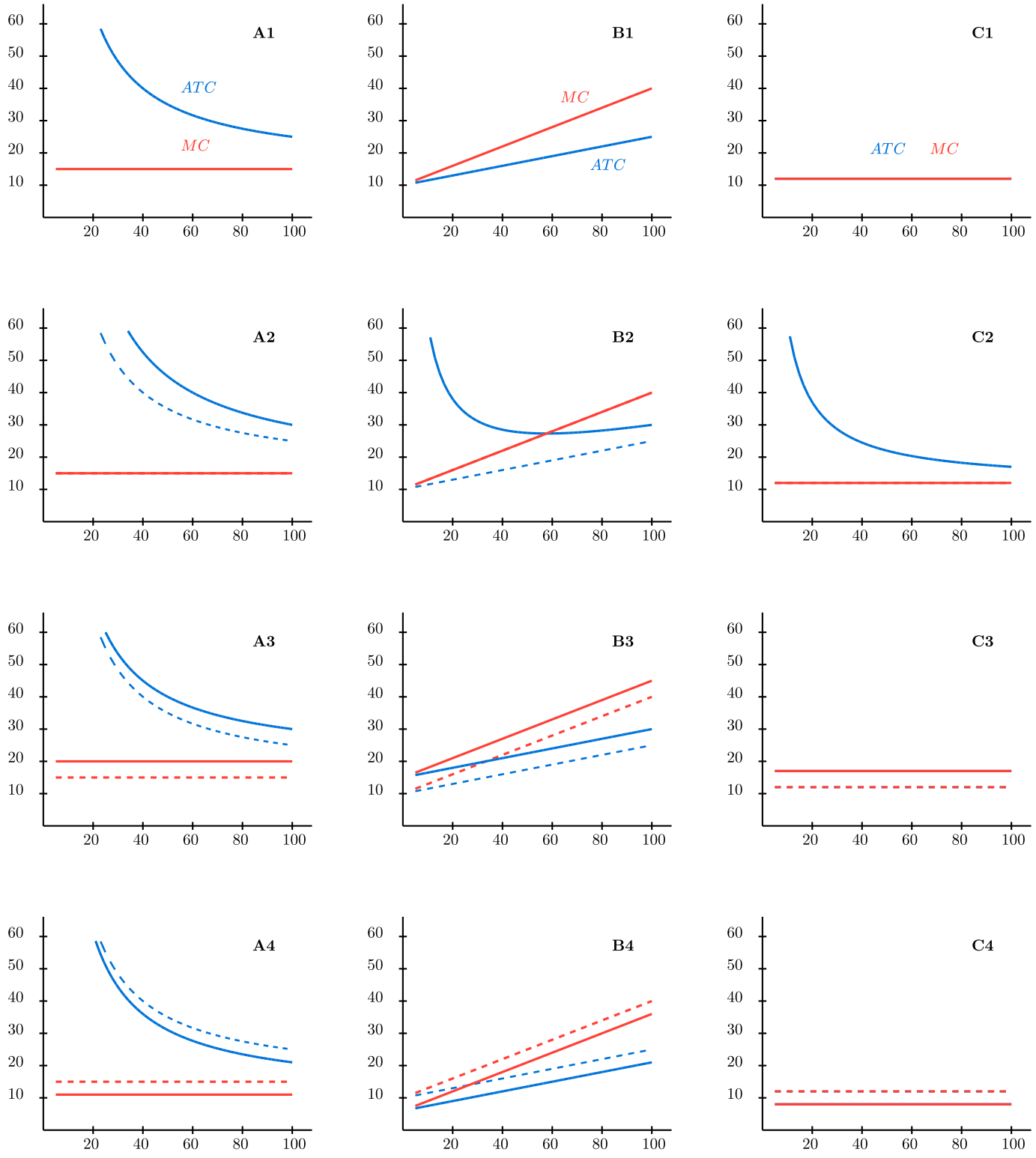
d) Lump Sum Tax

Which graph shows CloudSweep's cost curves after the Ministry imposes a 500 coin annual safety fee?

Graph A1 Graph A2 Graph A3 Graph A4

Graph A2. A fee that does not depend on output is a fixed cost. MC does not move. ATC rises by $500/Q$: by 12.5 at 40 brooms and by 5 at 100.

The graphs below show each manufacturer's ATC (blue) and MC (red). Row 1 is the base case. In Rows 2 to 4 the dashed curves repeat Row 1.


A1
ATC
MC
B1
MC
ATC
C1
ATC
MC
A2
B2
C2
A3
B3
C3
A4
B4
C4
60
50
40
30
20
10
20
40
60
80
100